

Informative Power-System Probing via Sequential Bayesian Experimental Design

Gaoming Lin
Texas A&M University
College Station, TX, USA
g.lin@tamu.edu

Byung-Jun Yoon
Texas A&M University
College Station, TX, USA
Brookhaven National Laboratory
Upton, NY, USA
bjyoon@tamu.edu

Abstract—To keep the power grid stable, sufficient knowledge of its latent parameters, such as inertia and frequency-response parameters, is important. Active probing is a practical approach to identify these parameters from system responses to injected probe signals, but conventional active probing is limited to fixed strategies that predefine all designs and cannot respond to new measurements, or to myopic ones that adapt only one step at a time and require high-load online optimization. To fill this gap, we regard active probing as a sequential Bayesian experimental design problem and apply adaptive methods with learned policies to maximize information about the system’s latent parameters. After experiments on different power grid systems, adaptive methods like deep adaptive design (DAD), Step-DAD, and reinforcement learning (RL-sBOED) achieve higher mean sPCE lower-bound information scores than myopic, optimized fixed-sequence, and random probing under the same probe budget. On the IEEE 9-bus model, DAD exceeds Fixed by 0.09–0.13 nats in mean sPCE score. Offline-trained DAD and RL-sBOED require less online decision time than Myopic and Step-DAD; on the IEEE 14-bus model with five probes, DAD takes 1.93 ms per sequence, compared with 16.94 s for Myopic and 44.69 s for Step-DAD under the recorded timing conditions. These results show the potential of learned sequential policies for informative and computationally effective adaptive probing in power grid systems.

Index Terms—Bayesian experimental design, power-system identification, adaptive probing, expected information gain, rate of change of frequency.

I. INTRODUCTION

As inverter-based resources replace synchronous generation, reduced physical inertia makes frequency-response characterization increasingly important [1]. Inertia mitigates rapid frequency changes, while frequency-response mechanisms adjust power in response to frequency deviations [2]. In the reduced models considered in this study, inertia M and effective frequency-response coefficients K represent these effects. These parameters are latent and must be inferred from observed system behavior. Reducing their uncertainty improves predictions of disturbance responses and supports informed control decisions.

Active probing is a well-established method for obtaining such information. In this approach, a small, known power signal is injected into the system, and the resulting frequency response is measured [2]. The observed response provides evidence for plausible parameter values; however, the information

gained depends on the probing-signal design. For instance, pulse duration influences the transient response and the quality of evidence available for parameter identification. This raises an experimental design question: how should a limited number of probes be selected to maximize learning about the latent parameters?

Bayesian optimal experimental design (BOED) serves as a formal framework for solving this question [3]. A prior distribution characterizes parameter uncertainty before experimentation. Bayes’ rule then combines this prior with the measurement model and observed response to yield a posterior distribution. Expected information gain (EIG) quantifies the anticipated average reduction in uncertainty resulting from an experiment, accounting for both latent parameters and measurement noise. EIG thus serves as a criterion for selecting a probe prior to observing its response.

For a sequence of experiments, fixed and myopic designs both present distinct limitations. On the one hand, a fixed sequence does not incorporate incoming measurements to update subsequent probes, so it fails to adapt to dynamic changes. On the other hand, a myopic design utilizes observation feedback but optimizes only the immediate next step’s EIG, neglecting future learning opportunities [3]. Additionally, repeated online optimization can make online decision-making computationally expensive. The central challenge is to integrate observation feedback with planning across the entire sequence while minimizing online computational costs.

Policy-based sequential BOED addresses this challenge. Early measurements inform subsequent choices, and training over complete sequences incorporates the value of future experiments. Deep adaptive design (DAD) [4] and reinforcement-learning-based sequential BOED (RL-sBOED) [5] learn a mapping from previous designs and observations to the next design. Offline training reduces optimization costs by reusing this mapping during experiments. Step-DAD [6] further refines a pretrained policy as new observations become available, balancing the value of additional adaptation against its online computational cost.

We apply these methods to probe-duration selection in reduced IEEE 9-bus and IEEE 14-bus models. These test networks contain nine and fourteen buses, respectively, which serve as electrical connection points for lines, generation, or

demand [7]. The reduced models preserve coupled machine dynamics, and the physical state is maintained between probes. We contribute a sequential grid-probing formulation and a systematic comparison of DAD, RL-sBOED, Step-DAD, Myopic, an optimized fixed sequence, and random probing. We assess the information gained under equal probe budgets and separately quantify offline training costs and online decision-making costs.

II. PROBLEM FORMULATION

We seek to reduce uncertainty about the power grid system through active probing injection within a budget of T design steps. Here $t = 1, \dots, T$ indexes design steps, d denotes elapsed time within each probe window, and D_t is the selected pulse duration.

A. Latent Parameters and Power-System Model

For a reduced network with m machines, the latent parameter vector is $\theta = (M_1, \dots, M_m, K_1, \dots, K_m)$, where M_i and K_i are inertia and effective frequency-response coefficients. The prior $p_0(\theta)$ has independent components $M_i \sim \mathcal{U}(0.7\bar{M}_i, 1.3\bar{M}_i)$ and $K_i \sim \mathcal{U}(0.1\bar{K}_i, \bar{K}_i)$. One draw of θ remains fixed throughout all T probes.

Let δ_i be rotor angle, ω_i angular-speed deviation, and u_i the applied probing power. The reduced swing model is

$$\begin{aligned} \dot{\delta}_i &= \omega_i, \\ M_i \dot{\omega}_i &= u_i - \left(D_i + \frac{K_i}{2\pi} \right) \omega_i \\ &\quad - \sum_{j=1}^m B_{ij} \sin(\delta_i - \delta_j). \end{aligned} \quad (1)$$

Dots denote derivatives with respect to physical time d . The damping D_i and network couplings B_{ij} are known; frequency deviation is $\Delta f_i = \omega_i/(2\pi)$ in Hz. The units of M_i , K_i and D_i are $\text{pu s}^2/\text{rad}$, pu/Hz and pu s/rad , respectively, on a 100-MVA base at nominal frequency 60 Hz. All angles and speed deviations start at zero. Net mechanical injections are zero in this deviation model, making the initial unforced state an equilibrium.

IEEE 9-bus retains buses 1, 2, and 3; IEEE 14-bus retains buses 1, 2, 3, 6, and 8. Test-case reactances [7] define a lossless Laplacian $\mathbf{L}$. Eliminating algebraic buses l while retaining machines g gives $\mathbf{L}_{\text{red}} = \mathbf{L}_{gg} - \mathbf{L}_{gl}\mathbf{L}_{ll}^{-1}\mathbf{L}_{lg}$, with $B_{ij} = -(\mathbf{L}_{\text{red}})_{ij}$ for $i \neq j$ and $B_{ii} = 0$. Resistance, voltage dynamics, and transformer tap effects are omitted. Table I summarizes the model parameterization. Nominal inertia is computed as $\bar{M}_i = 2H_i/(2\pi f_0)$, with $f_0 = 60$ Hz. For IEEE 9-bus, $H = (23.64, 6.40, 3.01)$ s matches published machine data [8, Table 2]. For IEEE 14-bus, $H = (4, 6.5, 5, 5, 5)$ s is prescribed at buses (1, 2, 3, 6, 8) as assumed benchmark values. For both networks, $\bar{K}_i = K_{\text{tot}} w_i$, with $K_{\text{tot}} = 0.5$ pu/Hz. The allocation weights are $w = (0.5875, 0.2625, 0.15)$ for IEEE 9-bus and $w = (8, 13, 10, 10, 10)/51$ for IEEE 14-bus. Fixed damping satisfies $D_i = (0.1 \text{ s}^{-1})\bar{M}_i$. The response

TABLE I
PRIOR SCALES AND DAMPING COEFFICIENTS

Network	Bus	$\bar{M}_i$	$\bar{K}_i$	D_i
IEEE 9-bus	1	0.1254141	0.2937500	0.0125414
	2	0.0339531	0.1312500	0.0033953
	3	0.0159685	0.0750000	0.0015969
IEEE 14-bus	1	0.0212207	0.0784314	0.0021221
	2	0.0344836	0.1274510	0.0034484
	3, 6, 8	0.0265257	0.0980392	0.0026526

scales, damping values, and prior ranges are benchmark assumptions rather than validated operational grid parameters.

B. Probe Design and Observation Model

The probe design is $\xi_t = (A_t, B_t, D_t)$: amplitude, bus location, and probe signal duration. This study fixes $A_t = 0.05$ pu and $B_t = 1$; only D_t is optimized. The raised-cosine input for $0 \leq d \leq D_t$ is

$$u_{B_t,t}(d) = \frac{A_t}{2} [1 - \cos(2\pi d/D_t)]. \quad (2)$$

The input is zero for $D_t < d \leq D_{\text{obs}}$ and at all other buses. The observation window is $D_{\text{obs}} = 3.5$ s. The full angle and speed state at its end becomes the initial state of the next window, without resetting the system. Consequently, the response depends on the complete design sequence $\xi_{1:t} = (\xi_1, \dots, \xi_t)$.

Write $\Delta f_{1,t}(d; \theta, \xi_{1:t})$ for the resulting bus-1 frequency deviation. The noise-free response is the signed endpoint rate of change of frequency (RoCoF), denoted by y_t :

$$y_t(\theta, \xi_{1:t}) = \frac{\Delta f_{1,t}(D_{\text{obs}}) - \Delta f_{1,t}(D_{\text{obs}} - \Delta d_{\text{samp}})}{\Delta d_{\text{samp}}}. \quad (3)$$

The frequency arguments $\theta, \xi_{1:t}$ are suppressed in the numerator. With sampling interval $\Delta d_{\text{samp}} = 0.025$ s, the measured response follows

$$y_{\text{obs},t} = y_t(\theta, \xi_{1:t}) + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma^2), \quad (4)$$

where $\sigma = 0.005$ Hz/s and noise is independent across windows. Each observation becomes available at window end. Decision latency is idealized as zero in simulated time.

C. Sequential Decisions and Bayesian Updating

Figure 1 summarizes the experiment-inference loop. The data after step t are the ordered design-observation history $h_t = ((\xi_k, y_{\text{obs},k}))_{k=1}^t$, with $h_0 = \emptyset$. A policy π selects ξ_t using only observed history h_{t-1} , with amplitude and bus fixed as above. It has no access to true parameters or physical state. Feasible durations satisfy

$$\begin{aligned} 0.2 &\leq D_t \leq 3, \quad t = 1, \dots, T, \\ D_t - D_{t-1} &\geq \Delta D_{\text{min}}, \quad t = 2, \dots, T, \end{aligned} \quad (5)$$

In this case, $\Delta D_{\text{min}} = 0.01$ s. At every step t , the upper boundary is defined as $3 - (T - t)\Delta D_{\text{min}}$ in order to leave enough room for the remaining probes. The lower boundary

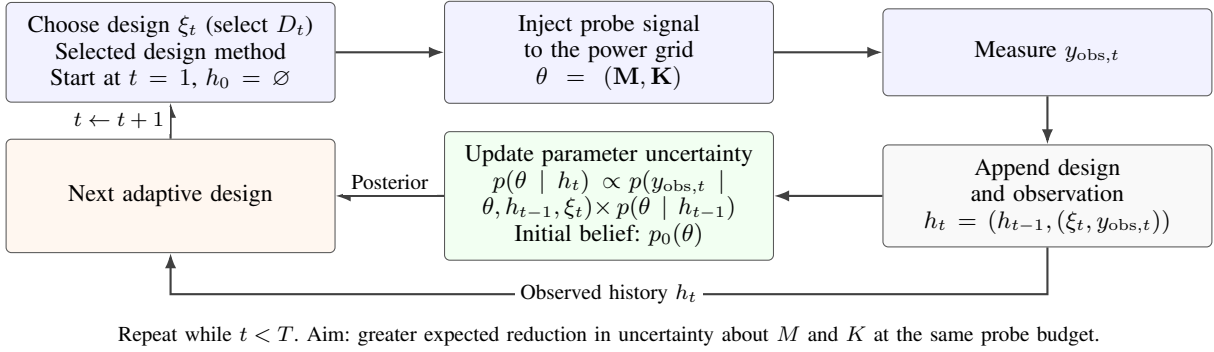


Fig. 1. Sequential probing with physical state carried between experiments. Observations update the design history and parameter uncertainty. For the next adaptive design, DAD/RL-sBOED use history directly, and Myopic/Step-DAD additionally maintain a particle posterior. Fixed and Random bypass observation feedback. The loop ends after T measurements.

starts at 0.2 and then becomes $D_{t-1} + \Delta D_{\min}$. All the methods convert a unit action into this interval without carrying out any sorting or discretization of the durations. Let Π denote the set of policies which meet these constraints; the policy randomness remains independent of θ .

The observation model is defined by a Gaussian likelihood $p(y_{\text{obs},t} | \theta, h_{t-1}, \xi_t)$, having a mean of $y_t(\theta, \xi_{1:t})$ and a variance of σ^2 ; Bayes' rule then updates the uncertainty about the parameters after each measurement has been taken:

$$p(\theta | h_t) = \frac{p(y_{\text{obs},t} | \theta, h_{t-1}, \xi_t) p(\theta | h_{t-1})}{p(y_{\text{obs},t} | h_{t-1}, \xi_t)}. \quad (6)$$

The denominator represents the probability density, through integrating the numerator over θ , where $p(\theta | h_0) = p_0(\theta)$. Each candidate parameter vector propagates its corresponding physical state under the observed designs to evaluate the likelihood. Design selection with history provides no additional information about θ ; so policy probabilities cancel from the Bayesian update.

D. Expected Information Gain and Design Objective

To compare possible design policies, we quantify the uncertainty reduction achieved by their observations using the Kullback–Leibler divergence between the terminal posterior and the prior. The expected information gain (EIG) of policy π is

$$\mathcal{I}_T(\pi) = \mathbb{E}_{h_T | \pi} [D_{\text{KL}}(p(\theta | h_T) \| p_0(\theta))]. \quad (7)$$

Here $D_{\text{KL}}(p \| q) = \int p(\theta) \log[p(\theta)/q(\theta)] d\theta$; information gain unit is nats. The expectation averages over parameter draws, measurement noise, and any policy randomness. The sequential Bayesian design problem is

$$\pi^* \in \arg \max_{\pi \in \Pi} \mathcal{I}_T(\pi). \quad (8)$$

Equation (8) defines the common information objective; the methods below differ in how they approximate or forgo its optimization. All receive the same number of probes. Computation is assessed separately.

EIG requires expensive marginal likelihoods. So, we use the sequential prior-contrastive (sPCE) lower bound [4] for training and evaluation, instead of the true EIG. Let $\theta^{(0)} \sim p_0$ generate the observed history and let $\theta^{(1)}, \dots, \theta^{(L)}$ be independent contrastive samples from the prior. For any candidate θ , define the observation likelihood along the realized sequence as

$$\mathcal{L}_t(\theta) = \prod_{k=1}^t p(y_{\text{obs},k} | \theta, h_{k-1}, \xi_k). \quad (9)$$

The sampled information score is then

$$\hat{I}_t = \log \frac{\mathcal{L}_t(\theta^{(0)})}{\frac{1}{L+1} \sum_{\ell=0}^L \mathcal{L}_t(\theta^{(\ell)})}. \quad (10)$$

Likelihoods are in the log domain after evaluation. Averaging $\hat{I}_T$ estimates the sPCE lower bound on $\mathcal{I}_T(\pi)$, not the exact EIG. Individual scores may be negative but are bounded above by $\log(L+1)$. With $L = 128$, the ceiling is 4.8598 nats; near-ceiling scores can conceal differences in EIG. We report these estimated lower-bound values as information scores throughout the results. The generating parameters are used only to score simulated observations.

III. COMPARED DESIGN METHODS

All six methods use the same observation model and feasible duration interval (5). Table II distinguishes observation feedback from greedy optimization: an adaptive method need not be myopic, and a fixed sequence can be optimized for the full horizon. The algorithms are adaptations of existing methods to the carried-state grid simulator.

A. Nonadaptive Designs: Random and Fixed

Random samples $z_t \sim \mathcal{U}(0,1)$ independently at each step and sets $D_t = D_t^{\min} + (D_t^{\max} - D_t^{\min})z_t$. Here $D_1^{\min} = 0.2$, $D_t^{\min} = D_{t-1} + \Delta D_{\min}$ for $t > 1$, and $D_t^{\max} = 3 - (T-t)\Delta D_{\min}$. Thus, random samples are uniform within the remaining feasible interval and are not uniform over feasible complete sequences. It does not consider observations and needs no offline training.

Fixed is a static Bayesian design: all durations are optimized jointly before observations are collected [3, Sec. 2.4]. The

TABLE II
COMPARISON OF EXPERIMENTAL DESIGN METHODS

Method	Adaptive	Myopic	Optimization
Random	No	N/A	None
Fixed [3]	No	No	Offline
Myopic [3]	Yes	Yes	Online
DAD [4]	Yes	No	Offline
RL-sBOED [5]	Yes	No	Offline
Step-DAD [6]	Yes	No	Both

ordered sequence is optimized using the full-sequence sPCE objective. It compares short, central, long and spread interior initializations using validation, then optimizes the selected initialization. Each training seed yields a separately fitted sequence. During evaluation, Fixed executes that sequence without adapting to observations.

B. Online One-Step Design: Myopic

Unlike Fixed, Myopic uses observation feedback. It follows one-step adaptive BOED, maximizing the expected information from the next observation under the current posterior [3, Sec. 2.4]. Our implementation maintains a weighted particle approximation to this posterior. Before each decision after the first, particles propagate their own states under the preceding duration and update likelihood weights using the latest observation. Continuous search then maximizes expected information from the next probe only, using 16 proposals, three search rounds, and 32 simulated observations per proposal. Candidate durations obey the same constraints as all other methods. No offline policy is trained.

C. Offline-Trained Adaptive Methods: DAD and RL-sBOED

DAD and RL-sBOED use neural policies trained on simulated sequences drawn from the same parameter prior as Fixed. Both methods receive the stage index and ordered trajectories, including past durations, scaled observations, and availability masks; future slots are masked. During evaluation, actions are generated via a forward pass without online optimization or explicit particle-posterior updates.

DAD [4] employs a deterministic two-layer, 64-unit tanh network, a trainable stage bias, and a sigmoid action map. Offline training maximizes sPCE by differentiating through the entire simulated sequence, including both observations and carried states. In joint Fixed/DAD training, validation determines whether to use random or interior Fixed-based initialization. Section V-B addresses the associated initialization cost.

RL-sBOED [5] has a REDQ-style ensemble actor-critic framework with ten critics and five replay updates per trajectory batch. The actor consists of two 128-unit ReLU layers. Per-step information rewards are defined as $R_t = \hat{I}_t - \hat{I}_{t-1}$, with $\hat{I}_0 = 0$ and a discount factor $\gamma = 1$. As a result, the cumulative reward will equal the terminal sPCE score. Offline training accounts for future design steps; online evaluation uses the trained actor without additional planning simulations.

Entropy is regularized by normalized tanh actions and a target entropy of -1 ; reported information scores do not include entropy bonuses. We use the actor’s deterministic output for evaluation.

D. Offline-Trained Online-Refined Adaptive Methods: Step-DAD

Step-DAD [6] evaluates whether adapting policy weights based on observed history yields improvements over offline training alone. Each episode starts with the corresponding selected DAD checkpoint, which remains unchanged for the initial probe. Prior to each subsequent decision, the latest observation is assimilated into a 128-particle posterior. The algorithm then samples candidate parameters and their associated carried states from this posterior to simulate the remaining design steps. These future simulations are conditioned on the actual observed history.

At each refinement step, a new Adam optimizer performs four gradient updates with a batch size of 32 and a learning rate of 10^{-3} . A fixed conditional validation batch is used to compare the unchanged policy with all updated candidates, and the best-performing candidate is retained. The retained weights are carried forward to the next design step within the same episode, but not across episodes. The total refinement budget is $4(T - 1)$ updates, corresponding to 8, 12, and 16 updates for $T = 3, 4, 5$, respectively.

For both Myopic and Step-DAD, assimilation is required only before subsequent decisions; neither planner requires a terminal update of the separately evaluated information score.

IV. EXPERIMENTAL SETTINGS

A. Simulation and Training Protocol

For each network, we use three, four, or five probes, making one observation for each probe. Although the various networks have the same sensor and design constraints, they differ in their physical parameters and policies.

The fourth-order Runge-Kutta (RK4) method is used to integrate the swing equations, with a step size of $1/640$ s. The numerical state and duration Jacobians are calculated using perturbations of 10^{-5} and take into account the derivatives of the carried states. The validation suite compares the policy gradients and the conditional refinement gradients with those obtained by end-to-end finite differences.

Offline training involves 2000 updates, using a batch size of 32 and three separate training seeds (101, 202, and 303); the DAD and fixed learning rates are both set to 10^{-3} . Validation is carried out using 128 systems every 100 updates. For the sPCE objective, 128 contrastive samples are used in the training, validation, and evaluation phases. The Myopic and Step-DAD methods each use 128 particles for online posterior planning, and Step-DAD carries out four refinement updates after each observation so long as design steps remain. The evaluation comprises seeds 1001, 1002, and 1003, with 512 simulated experimental sequences per evaluation seed. Validation checkpoints include no gradient updates for the validation cases. Each trained policy is assessed over 1536

episodes. The draws for parameters and measurement noise are paired across the different methods, whereas the online planner makes use of its own separate random stream. Since the Random and Myopic methods show no dependence on the training seed under these streams, identical evaluations can be obtained for all training seeds. Fixed is optimized independently for each training seed.

B. Evaluation and Computational Measurements

With respect to the trained methods, the evaluation-seed means are averaged for each training seed, and the mean $\pm$ sample standard deviation is given based on three trainings. In the case of the Random and Myopic methods, it is the mean $\pm$ sample standard deviation of the evaluation-seed means that is reported, not the dispersion at the episode level. Evaluation seed 1001 was used during development, so the evaluation is not fully independent.

The experiments use NVIDIA RTX 5080 and A100 GPUs, with the RTX 4090 used for development. When combining the results from IEEE 9-bus for offline processing, both the RTX 5080 and A100 are taken into account, whereas for the online timings only the RTX 5080 is used, except in the case of Step-DAD when $T=5$ and the training seed is 303, which was run on an A100. For IEEE 14-bus, the timings are based on the A100. The workflow costs that have been recorded do not allow for a controlled comparison of the run times between different networks. Code and saved results are available at https://github.com/alexschmidt123/tpec2027_power_grid.

V. RESULTS AND DISCUSSION

A. Information Acquired at Equal Probe Budgets

Tables III and IV report mean sPCE information scores in nats, with the seed-level standard deviations defined above. All three learned adaptive methods have higher mean sPCE scores than Random, Fixed, and Myopic in the reported network-horizon comparisons.

TABLE III
IEEE 9-BUS INFORMATION SCORE (SPCE, NATS)

Method	$T = 3$	$T = 4$	$T = 5$
Random	0.4321 ± 0.0310	0.4924 ± 0.0342	0.5392 ± 0.0471
Fixed	2.0112 ± 0.0116	2.3192 ± 0.0004	2.5066 ± 0.0013
Myopic	1.9736 ± 0.0438	2.1707 ± 0.0366	2.2407 ± 0.0547
DAD	2.1366 ± 0.0161	2.4095 ± 0.0080	2.6351 ± 0.0179
RL-sBOED	2.1257 ± 0.0027	2.3963 ± 0.0333	2.5307 ± 0.0114
Step-DAD	2.1471 ± 0.0110	2.4382 ± 0.0048	2.6621 ± 0.0145

On IEEE 9-bus, DAD exceeds Fixed by 0.1253, 0.0903, and 0.1286 nats at $T = 3, 4, 5$, respectively. Step-DAD improves the DAD mean by 0.0105, 0.0287, and 0.0269 nats. These mean improvements are small; paired uncertainty estimates are needed to assess their reliability. The IEEE 9-bus information scores are well below the estimator ceiling.

On IEEE 14-bus, DAD exceeds Myopic by 0.4669 nats and Fixed by 0.4953 nats at $T = 3$. Fixed exceeds Myopic at $T =$

TABLE IV
IEEE 14-BUS INFORMATION SCORE (SPCE, NATS)

Method	$T = 3$	$T = 4$	$T = 5$
Random	0.7541 ± 0.0047	0.8360 ± 0.0297	0.8464 ± 0.0252
Fixed	4.2623 ± 0.0402	4.4946 ± 0.0345	4.6624 ± 0.0083
Myopic	4.2907 ± 0.0368	4.3990 ± 0.0177	4.4641 ± 0.0280
DAD	4.7576 ± 0.0362	4.8325 ± 0.0047	4.8443 ± 0.0026
RL-sBOED	4.7445 ± 0.0022	4.8275 ± 0.0112	4.8383 ± 0.0108
Step-DAD	4.7527 ± 0.0275	4.8297 ± 0.0067	4.8446 ± 0.0042

4, 5. This result shows that observation feedback alone does not guarantee a more informative design when optimization is myopic. The adaptive learned methods combine feedback with a full-sequence objective, although this comparison does not isolate the contribution of each mechanism.

Differences among learned policies on IEEE 14-bus are small: Step-DAD changes the DAD mean by -0.0049 , -0.0028 , and $+0.0003$ nats at $T = 3, 4, 5$. At $T = 5$, DAD is only 0.0155 nats below the 4.8598-nat ceiling. The results therefore show no consistent benefit from refinement and cannot establish a reliable ranking of the learned policies' exact EIG.

B. Measured Computational Cost

Tables V through VIII list the offline training time (in hours) and online decision-making time (in milliseconds) for each complete sequence, respectively. Online timing sums all T decisions, including posterior updates, Myopic search, and Step-DAD refinement where applicable. It excludes physical response simulation, observation acquisition, and information scoring. For each evaluation seed, we report the average time over 512 episodes and, where applicable, over training seeds; the reported mean and sample SD use the three evaluation-seed means.

TABLE V
IEEE 9-BUS OFFLINE STAGE TIME (HOURS)

Method	$T = 3$	$T = 4$	$T = 5$
Random	0	0	0
Fixed	0.960 ± 0.352	1.396 ± 0.515	1.827 ± 0.662
Myopic	0	0	0
DAD	0.957 ± 0.342	1.393 ± 0.509	1.831 ± 0.654
RL-sBOED	0.120 ± 0.017	0.152 ± 0.030	0.180 ± 0.045
Step-DAD	0.957 ± 0.342	1.393 ± 0.509	1.831 ± 0.654

Offline timing includes initialization, optimization, periodic validation, and checkpoint writes. The DAD entries report its training stage only and exclude the separately incurred Fixed initialization cost; the total Fixed-DAD workflow requires both stages. Step-DAD makes use of the same matching DAD model: its offline entries repeat the DAD stage cost without a separate offline training run; the refinement costs are incurred online. For the offline means and the sample standard

TABLE VI
IEEE 14-BUS OFFLINE STAGE TIME (HOURS)

Method	$T = 3$	$T = 4$	$T = 5$
Random	0	0	0
Fixed	1.877 ± 0.011	2.750 ± 0.002	3.691 ± 0.050
Myopic	0	0	0
DAD	1.894 ± 0.032	2.725 ± 0.018	3.626 ± 0.068
RL-sBOED	0.170 ± 0.008	0.204 ± 0.001	0.248 ± 0.001
Step-DAD	1.894 ± 0.032	2.725 ± 0.018	3.626 ± 0.068

TABLE VII
IEEE 9-BUS ONLINE DECISION TIME (MS PER SEQUENCE)

Method	$T = 3$	$T = 4$	$T = 5$
Random	0.0715 ± 0.0001	0.09445 ± 0.00003	0.121 ± 0.001
Fixed	0.523 ± 0.038	0.623 ± 0.002	0.766 ± 0.002
Myopic	9308 ± 4	12577 ± 2	15892 ± 2
DAD	0.537 ± 0.002	0.683 ± 0.002	0.838 ± 0.004
RL-sBOED	0.492 ± 0.003	0.626 ± 0.001	0.764 ± 0.002
Step-DAD	6342 ± 2	15808 ± 9	24350 ± 50

TABLE VIII
IEEE 14-BUS ONLINE DECISION TIME (MS PER SEQUENCE)

Method	$T = 3$	$T = 4$	$T = 5$
Random	0.161 ± 0.004	0.209 ± 0.002	0.309 ± 0.049
Fixed	1.16 ± 0.14	1.48 ± 0.04	2.75 ± 1.73
Myopic	10000 ± 2	13437 ± 9	16935 ± 35
DAD	1.22 ± 0.09	1.42 ± 0.02	1.93 ± 0.20
RL-sBOED	1.68 ± 0.29	3.31 ± 1.33	2.71 ± 0.69
Step-DAD	9113 ± 7	23243 ± 24	44693 ± 32

deviations, three training seeds are used. Since Random and Myopic need no offline training, their entries are zero.

At $T = 5$, DAD needs 1.93 ms per sequence online, whereas Myopic requires 16.94 s and Step-DAD needs 44.69 s (as shown in Table VIII). Although the DAD and Step-DAD scores are nearly the same, their computational costs are very different. The advantage obtained through offline training consists in spreading the cost of the design computation over a number of repeated experiments, but this benefit depends on how frequently the trained policy is reused and on the time available between measurements.

C. Optimization and Estimator Diagnostics

Evaluation uses validation-selected checkpoints. We cannot guarantee that equal convergence or equal computational effort will be achieved within the same update budget since DAD initialization also makes use of a separately trained fixed sequence.

D. Scope and Limitations

These conclusions pertain to the reduced lossless dynamics, the independent parameter priors, and the scalar sensor as

specified in this case. The same parameter distribution is used during both training and evaluation; we have not demonstrated robustness with respect to prior shift, sensor bias, or unmodelled dynamics. The evaluation seed 1001 was used to guide the development process, and therefore the overall results do not constitute completely independent confirmation. The errors introduced by finite-contrast scoring and particle inference are different from those caused by ODE integration. Lastly, the zero-latency simulation does not prove that online planners are feasible in real time, and increased information gain on its own does not show that control is safer.

VI. CONCLUSION

Adaptive methods based on learned policies use the observation history to optimize the selection throughout the entire experimental sequence. When three to five probes are used on the reduced IEEE 9-bus and 14-bus models, DAD, RL-sBOED, and Step-DAD obtain higher mean sPCE information scores than Random, Fixed, and Myopic. Estimator saturation limits comparisons among learned policies on the IEEE 14-bus model. DAD and RL-sBOED, trained offline, amortize the cost of online decisions, while Step-DAD incurs a considerably higher cost even though it provides only modest additional gains on the IEEE 9-bus model and shows no consistent improvement on the IEEE 14-bus model. Before concluding that these methods are ready for deployment, future work should evaluate estimator sensitivity, model mismatch, and decision latency.

REFERENCES

- [1] P. Denholm, T. Mai, R. W. Kenyon, B. Kroposki, and M. O'Malley, "Inertia and the power grid: A guide without the spin," National Renewable Energy Laboratory, Tech. Rep. NREL/TP-6A20-73856, 2020. [Online]. Available: <https://docs.nrel.gov/docs/fy20osti/73856.pdf>
- [2] J. Peng, J. Tan, P. Koralewicz, E. Mendiola, S. Dong, A. Hoke, and K. Horowitz, "Probing signal-based inertia and frequency response estimation for power systems with high levels of inverter-based resources," in *2024 IEEE Power & Energy Society General Meeting (PESGM)*, 2024, pp. 1–5.
- [3] X. Huan and Y. M. Marzouk, "Sequential Bayesian optimal experimental design via approximate dynamic programming," arXiv:1604.08320, 2016. [Online]. Available: <https://arxiv.org/abs/1604.08320>
- [4] A. Foster, D. R. Ivanova, I. Malik, and T. Rainforth, "Deep adaptive design: Amortizing sequential Bayesian experimental design," in *Proceedings of the 38th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 139, 2021, pp. 3384–3395.
- [5] T. Blau, E. V. Bonilla, I. Chades, and A. Dezfouli, "Optimizing sequential experimental design with deep reinforcement learning," in *Proceedings of the 39th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 162, 2022, pp. 2107–2128.
- [6] M. Hedman, D. R. Ivanova, C. Guan, and T. Rainforth, "Step-DAD: Semi-amortized policy-based Bayesian experimental design," in *Proceedings of the 42nd International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 267, 2025, pp. 22904–22923. [Online]. Available: <https://proceedings.mlr.press/v267/hedman25a.html>
- [7] R. D. Zimmerman, C. E. Murillo-Sanchez, and R. J. Thomas, "MAT-POWER: Steady-state operations, planning, and analysis tools for power systems research and education," *IEEE Transactions on Power Systems*, vol. 26, no. 1, pp. 12–19, 2011.
- [8] R. Kaur and D. Kumar, "Transient stability improvement of IEEE 9 bus system using power world simulator," *MATEC Web of Conferences*, vol. 57, p. 01026, 2016.